Xinyi (Cynthia) Li

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EDUCATION

Carnegie Mellon University, Tepper School of Business — Pittsburgh, PA
B.S. in Economics and Mathematical Sciences — Aug. 2024 – June 2027 (Expected)

- GPA: 3.95
- Coursework: Real Analysis I & II, Complex Analysis (graduate), Modern Regression, Probability and Statistical Inference I & II, Matrices and Linear Transformations, Discrete Mathematics, Differential Equations, Fundamentals of Programming and Computer Science, Measure and Integration (graduate, in progress), Linear algebra (in progress)
- Planned (Spring 2027): Advanced Methods for Data Analysis, Introduction to Probability Modeling

The University of Hong Kong, Business school — Hong Kong
B.S. in Quantitative Finance (transferred to CMU) — Sept. 2023 – June 2024

- GPA: 3.96
- Coursework: Multivariable Calculus and Linear Algebra, University Mathematics II, Statistics: Ideas and Concepts

RESEARCH EXPERIENCE

Research Assistant, Department of Statistics, CMU — Oct. 2025 – Present
Supervised by Prof. Nynke M. D. Niezink · Local network autocorrelation

- Designed and led a study adapting local autocorrelation measures (local Moran's I and local join count statistics) from spatial statistics to social networks to capture individual-level variation in attribute similarity that global network autocorrelation measures miss.
- Conducted stochastic actor-oriented model (SAOM) simulations to test how local Moran's I responds to homophilic selection and social influence, and permutation analyses that randomize either the network or the attribute to test whether local Moran's I is sensitive to changes in network structure and behavior.
- Applied local autocorrelation measures to two longitudinal friendship datasets covering stress, alcohol use, and smoking, and added local average similarity as an additional measure, identifying gender differences in local clustering
- First-author manuscript submitted to *Sociological Methods & Research* (see Manuscripts), with a reproducible code package covering all simulations and empirical examples.

Research Assistant, Department of Statistics, CMU — Jul. 2026 – Present
Supervised by Prof. Nynke M. D. Niezink · Multiplex p2 models for networks

- Extend the Bayesian multiplex p2 model to jointly accommodate directed and undirected network layers (e.g., advice seeking and collaboration), which existing p_2 models cannot handle.
- Derive the dyad-level likelihood for mixed multiplex networks, accounting for joint configurations and cross-layer dependence of directed and undirected ties.
- Reproduced published directed multiplex p2 applications to the Lazega law firm and ChemNet policy networks, assessing model fit through posterior predictive checks
- Implement the undirected-layer extension in the R package **multip2** and extend the package to fit mixed directed–undirected network models

Research Assistant, Department of Statistics, University of Pittsburgh — Sept. 2026 – Present
Supervised by Prof. Jiebiao Wang · Spatial inference, Alzheimer's disease spatial proteogenomics

- Lead spatial analyses of single-cell spatial transcriptomics data to investigate how cell–cell and cell–pathology spatial relationships vary with Alzheimer's disease progression.
- Implemented KAMP-based spatial inference to assess cell-type colocalization while accounting for tissue inhomogeneity.
- Applied functional data analysis to summarize colocalization across spatial scales and assess associations with disease stage, adjusting for donor-level covariates.
- Proposed and implemented a network-based approach to assess global and local cell-type adjacency patterns in brain tissue

Research Assistant, Department of Statistics, CMU — Mar. 2025 – Aug. 2025
Supervised by Prof. Weijing Tang · Item response theory for evaluating large language models

- Extended IRT-based LLM evaluation to multidimensional latent abilities with embedding-based item parameters, enforcing identifiability through centering, scale constraints, and Procrustes alignment.
- Implemented and compared maximum likelihood estimation and Monte Carlo EM, with and without embeddings, in simulation studies across varying numbers of items.

- Demonstrated that question embeddings improved item-parameter estimation under both estimation approaches

Research Assistant, Department of Statistics & Data Science, CMU Aug. 2026 – Present
- Score student statistics reports and meet with other graders to discuss scoring differences and improve consistency

MANUSCRIPTS

- Li, C. X. and Niezink, N. M. D. *Local Network Autocorrelation: Uncovering Individual Heterogeneity in Attribute Clustering.* Submitted to Sociological Methods & Research.

PRESENTATIONS

- Poster, 2026 Conference on Statistical Learning and Data Science (SLDS), New York Nov. 2026
- Local network autocorrelation, biostatistics research group, Upitts Oct. 2026
- Local network autocorrelation, Meeting of the Minds Undergraduate Research Symposium, CMU May 2026
- Local network autocorrelation, CMU Networkshop Apr. 2026

AWARDS AND FUNDING

- Summer Undergraduate Research Fellowship (SURF), Carnegie Mellon University Summer 2026
- Summer Undergraduate Research Apprenticeship (SURA), Carnegie Mellon University Summer 2025

ADDITIONAL INFORMATION

- Technical skills: R, Python, SQL, LaTeX, Tableau, HTML
- Languages: Mandarin Chinese (native), English (fluent)
- Interests: theater, poetry writing (Mandarin), pipa, reading, hiking